

No Privacy for Privates: Veterans Prolific Survey

Start of Block: Consent

Q1

RESEARCH PARTICIPANT Information Sheet

Privacy Risks of Mobile Apps Marketed Toward Military-Affiliated Personnel

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- You are being asked to participate or be a part of a research study. Your participation is **voluntary** which means that you may choose not to participate at any time.
- The researchers hope to learn more about how U.S. military affiliated adults view the privacy and security risks of mobile apps marketed toward military affiliated personnel.
- You are being asked to participate because you are 18 years or older and are affiliated with the U.S. military (e.g., active duty, Reserve/National Guard, veteran, an adult family member, or DoD civilian).
- You will be asked to take a survey about what mobile app data you consider sensitive, what kinds of attacks you think are possible if that data is collected, and how well your current security practices and guidance address those risks.
- The study will take a total of 30-60 minutes.

Please take time to review the rest of the information. This will give you information about this study to help you decide if you want to participate.

This study is only intended for people who are 18 years of age or older. Do not complete the study if you are not legally considered an adult.

Before agreeing to participate, consider the risks and potential benefits of taking part in this study. All research carries the risk of breach of confidentiality which means that someone outside of our study could figure out that you were in the study or information was yours. How we will protect your information to reduce this risk is below.

Some questions could make you feel uncomfortable. You can skip any of these questions or stop answering.

It is unlikely that there will be personal benefits to you for participating. Having more information from the answers in this research study might help us or other researchers understand more about how U.S. military affiliated adults view the privacy and security risks of mobile apps marketed toward military affiliated personnel.

Will I receive payment or other incentive? To thank you for being in our research study, if you were recruited through Prolific, immediately after completing the survey, you will receive \$10 through Prolific's payment method. If you were recruited through open recruitment, after you complete the survey, you will receive a \$10 Amazon gift card to the email address you provide at the end of the survey.

"According to the rules of the Internal Revenue Service (IRS), payments that are made to you as a result of your participation in a study may be considered taxable income."

How will the researchers protect my information, privacy, and confidentiality? Your personal information may be shared outside the research study if required by law. We also may need to share your research records with other groups for quality assurance or data analysis. These groups include the Purdue University Institutional Review Board or its designees, and state or federal agencies who may need to access the research records (as allowed by law). Your research information may also be shared with the study sponsor, United States Military Academy (USMA) at West Point.

The study team plans to keep answers for this study to answer research questions. We will keep this information until we are done with the study, approximately 1 year and for at least three years after we are finished.

We may share the anonymous data and findings with other researchers or in research papers or presentations. We may include direct quotes from your open-ended survey responses in publications, presentations, or articles. Any quotes used will be reviewed to ensure they do not contain personally identifying information or details that could identify you or your specific workplace/unit.

What are my rights as a research participant in this study? You do not have to participate in this research project. If you agree to participate, you may withdraw your participation at any time without penalty.

For participants not recruited through Prolific, your decision to participate or not participate in this research will have no effect on your relationship with your employer, military unit, or any organization.

Who can I contact if I have questions about the study? If you have questions, comments or concerns about this research project, you can talk to one of the researchers.

Please primarily contact Antonio Bianchi at antonioab@purdue.edu (secondarily 7654947100) or secondarily contact Joshua Shinkle at jcshinkl@purdue.edu.

To report anonymously via Purdue's Hotline, see www.purdue.edu/hotline

If you have questions about your rights while taking part in the study or have concerns about the treatment of research participants, please call the Human Research Protection Program at (765) 494-5942, email (irb@purdue.edu) or write to:

Human Research Protection Program

Purdue University
Seng Liang Wang Hall
516 Northwestern Ave,
West Lafayette, IN 47906

By answering “I agree” to the question below, I agree to take part in this research. **I am 18 years of age or older and understand the information above about my participation.**

Q2 Do you agree to the consent form?

- ☐ I agree
- ☐ I do not agree

End of Block: Consent

Start of Block: Pre-screening

Q3 Are you 18 years or older?

- ☐ Yes
 - ☐ No
-

Q4 Are you an active military service member?

- ☐ I'm an active US military service member
 - ☐ I'm an active UK military service member
 - ☐ I'm an active military service member from another country
 - ☐ A member of my family is an active military member
 - ☐ No, I am not
-

Q5 Are you a military services veteran?

- ☐ I'm a US services veteran
 - ☐ I'm a UK services veteran
 - ☐ I'm a services veteran from another country
 - ☐ No
-

Q6 Are you in the US Reserve / National Guard?

- ☐ I'm in the US Reserve
 - ☐ I'm in the US National Guard
 - ☐ No
-

Q7 Are you the family member of an active US military service member, a US services veteran, or a US Reserve/National Guard member?

- ☐ I'm the family member of an active US military service member
 - ☐ I'm the family member of a US services veteran
 - ☐ I'm the family member of a US Reserve/National Guard member
 - ☐ No
-

Q8 Are you a DoD civilian employee?

- ☐ I'm a DoD civilian employee
- ☐ No

End of Block: Pre-screening

Start of Block: Device/App Usage

Q9 In your everyday life, as a smartphone user, do you primarily use a personal device, a government/military-issued device, or a mix of both?

- ☐ Personal Device
 - ☐ Government/Military-Issued Device
 - ☐ Mix of Both
-

Q10 Is the device you use most often an Android or an iPhone?

- ☐ Android
 - ☐ iPhone
-

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Q11 The following apps market themselves specifically toward military-affiliated personnel. We define military-affiliated personnel as anyone who is an active US military service member, a US services veteran, a US Reserve / National Guard member, a family member of an active US military service member/US services veteran/US Reserve/National Guard member, or a DoD civilian employee.

Q12 For the following apps, indicate how frequently you or a family member use it in daily life.

	Always	Often	Sometimes	Rarely	Never
Hots&Cots	☐	☐	☐	☐	☐
Ace PDG+SJT 2026 USAF PFE Prep	☐	☐	☐	☐	☐
AFI Explorer	☐	☐	☐	☐	☐
Bluejacketeer	☐	☐	☐	☐	☐
USAF Connect	☐	☐	☐	☐	☐
MyNavy UNIFORMS	☐	☐	☐	☐	☐
USAA Mobile	☐	☐	☐	☐	☐
Navy Federal Credit Union	☐	☐	☐	☐	☐
Official Navy PFA	☐	☐	☐	☐	☐
My Military OneSource	☐	☐	☐	☐	☐
ASVAB Mastery: ASVAB Test	☐	☐	☐	☐	☐
MGRS UTM GPS	☐	☐	☐	☐	☐
Mgrs & Utm Map	☐	☐	☐	☐	☐
ATAK-CIV (Civil Use)	☐	☐	☐	☐	☐
VA: Health and Benefits	☐	☐	☐	☐	☐

Sandboxx	☐	☐	☐	☐	☐
VET Tv	☐	☐	☐	☐	☐
Aim High Air Force	☐	☐	☐	☐	☐
Armed Forces Bank	☐	☐	☐	☐	☐
ASVAB Flashcards	☐	☐	☐	☐	☐
ASVAB Practice for Dummies	☐	☐	☐	☐	☐
ASVAB Practice Test 2024 Prep	☐	☐	☐	☐	☐
CBT-i Coach	☐	☐	☐	☐	☐
Defense & Military News	☐	☐	☐	☐	☐

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Q13 Are there any apps not previously listed that you or a family member use because of your military affiliation?

☐ Yes

☐ No

Q14 Please list them.

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Q15 For each of the following app categories, what data/permissions do you provide when using mobile apps from that category? Select all that apply.

	Finance (e.g., USAA Mobile)	Health (e.g., PTSD Coach)	Exam Prep (e.g., ASVAB Flashcards)	Installation Resources (e.g., Hots&Cots)	Geospatial Tools (e.g., Land Nav Assistant)	Media (e.g., VET Tv)	Social (e.g., MilitaryCupid: Military Dating)	Career (e.g., U.S. Army Career Navigator)
Email Address	☐	☐	☐	☐	☐	☐	☐	☐
Name	☐	☐	☐	☐	☐	☐	☐	☐
Phone Number	☐	☐	☐	☐	☐	☐	☐	☐
Military Base / Installation	☐	☐	☐	☐	☐	☐	☐	☐
Branch	☐	☐	☐	☐	☐	☐	☐	☐
Address	☐	☐	☐	☐	☐	☐	☐	☐
DOB	☐	☐	☐	☐	☐	☐	☐	☐
Gender	☐	☐	☐	☐	☐	☐	☐	☐
Sexual Orientation	☐	☐	☐	☐	☐	☐	☐	☐
Age	☐	☐	☐	☐	☐	☐	☐	☐
Financial Information	☐	☐	☐	☐	☐	☐	☐	☐
App Activity	☐	☐	☐	☐	☐	☐	☐	☐
Access Camera	☐	☐	☐	☐	☐	☐	☐	☐

Access Location Only While Using App	☐	☐	☐	☐	☐	☐	☐	☐
Access Background Location	☐	☐	☐	☐	☐	☐	☐	☐
Make / Manage Phone Calls	☐	☐	☐	☐	☐	☐	☐	☐
Record Audio	☐	☐	☐	☐	☐	☐	☐	☐
Access Contacts	☐	☐	☐	☐	☐	☐	☐	☐
Access Health and Fitness Data	☐	☐	☐	☐	☐	☐	☐	☐
Access SMS / iMessages	☐	☐	☐	☐	☐	☐	☐	☐
☒ None of the above	☐	☐	☐	☐	☐	☐	☐	☐

End of Block: Device/App Usage

Start of Block: Perception of Data Sensitivity

Q16 Rank the following data points and permissions from the one that makes you most uncomfortable when collected (top) to the one that makes you least uncomfortable when collected (bottom).

- _____ Email Address
- _____ Name
- _____ Phone Number
- _____ Military Base / Installation
- _____ Branch
- _____ Address
- _____ DOB
- _____ Gender
- _____ Sexual Orientation
- _____ Age
- _____ Financial Information
- _____ App Activity
- _____ Device Identifiers
- _____ Access Camera
- _____ Access Approximate Location
- _____ Access Fine Location
- _____ Access Background Location
- _____ Make / Manage Phone Calls
- _____ Record Audio
- _____ Access Contacts
- _____ Access Health and Fitness Data
- _____ Access SMS / iMessages

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Q17 How comfortable are you with a mobile app collecting each of the following data points from you?

	Extremely uncomfortable	Somewhat uncomfortable	Neither comfortable nor uncomfortable	Somewhat comfortable	Extremely comfortable
Email Address	☐	☐	☐	☐	☐
Name	☐	☐	☐	☐	☐
Phone Number	☐	☐	☐	☐	☐
Military Base / Installation	☐	☐	☐	☐	☐
Branch	☐	☐	☐	☐	☐
Address	☐	☐	☐	☐	☐
DOB	☐	☐	☐	☐	☐
Gender	☐	☐	☐	☐	☐
Sexual Orientation	☐	☐	☐	☐	☐
Age	☐	☐	☐	☐	☐
Financial Information	☐	☐	☐	☐	☐
App Activity	☐	☐	☐	☐	☐
Device Identifiers	☐	☐	☐	☐	☐
Permission: Access Camera	☐	☐	☐	☐	☐

Permission: Access Approximate Location	☐	☐	☐	☐	☐
Permission: Access Fine Location	☐	☐	☐	☐	☐
Permission: Access Background Location	☐	☐	☐	☐	☐
Permission: Make / Manage Phone Calls	☐	☐	☐	☐	☐
Permission: Record Audio	☐	☐	☐	☐	☐
Permission: Access Contacts	☐	☐	☐	☐	☐
Permission: Access Health and Fitness Data	☐	☐	☐	☐	☐
Permission: Access SMS / iMessages	☐	☐	☐	☐	☐

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Q18 Compared with your answers from the previous question, would you feel more, less, or equally uncomfortable if that data was...(a) collected by a mobile app marketed toward military-affiliated personnel, (b) shared with a foreign party, or (c) shared with an unknown third party? Why? **Note: Treat each scenario independently.**

Collected by an App Marketed Towards Military-Affiliated Personnel			Shared with a Foreign Party			Shared with a Third-Party			Justification
More uncomfortable	Equally uncomfortable	Less uncomfortable	More uncomfortable	Equally uncomfortable	Less uncomfortable	More uncomfortable	Equally uncomfortable	Less uncomfortable	Why?

Email Address	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Name	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Phone Number	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Military Base / Installation	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Branch	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Address	☐	☐	☐	☐	☐	☐	☐	☐	☐	
DOB	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Gender	☐	☐	☐	☐	☐	☐	☐	☐	☐	

Sexual Orientation	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Age	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Financial Information	☐	☐	☐	☐	☐	☐	☐	☐	☐	
App Activity	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Device Identifiers	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Permission: Access Camera	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Permission: Access Approximate Location	☐	☐	☐	☐	☐	☐	☐	☐	☐	
Permission: Access Fine Location	☐	☐	☐	☐	☐	☐	☐	☐	☐	

Permission: Access Background Location	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Permission: Make / Manage Phone Calls	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Permission: Record Audio	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Permission: Access Contacts	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Permission: Access Health and Fitness Data	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Permission: Access SMS / iMessages	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

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Q19 For the following countries listed in the 2023 DoD Cyber Strategy, how comfortable would you be using an app that included code/SDKs/trackers from that country? Why?

	Response					Justification
	Extremely uncomfortable	Somewhat uncomfortable	Neither comfortable nor uncomfortable	Somewhat comfortable	Extremely comfortable	Why?
People's Republic of China	☐	☐	☐	☐	☐	
Russia	☐	☐	☐	☐	☐	
North Korea	☐	☐	☐	☐	☐	
Iran	☐	☐	☐	☐	☐	

Q20 Compared with your answers from the previous question, would you feel more, less, or equally uncomfortable if you or your family member(s) were not affiliated with the military?

	More uncomfortable	Equally uncomfortable	Less uncomfortable
People's Republic of China	☐	☐	☐
Russia	☐	☐	☐
North Korea	☐	☐	☐
Iran	☐	☐	☐

Q21 For each app listed, do you believe it contains code from any of the following countries:
People's Republic of China, Russia, North Korea, Iran?

	Yes	Maybe	No
Hots&Cots	☐	☐	☐
Ace PDG+SJT 2026 USAF PFE Prep	☐	☐	☐
AFI Explorer	☐	☐	☐
Bluejacketeer	☐	☐	☐
USAF Connect	☐	☐	☐
MyNavy UNIFORMS	☐	☐	☐
USAA Mobile	☐	☐	☐
Navy Federal Credit Union	☐	☐	☐
Official Navy PFA	☐	☐	☐
My Military OneSource	☐	☐	☐
ASVAB Mastery: ASVAB Test	☐	☐	☐
MGRS UTM GPS	☐	☐	☐
Mgrs & Utm Map	☐	☐	☐
ATAK-CIV (Civil Use)	☐	☐	☐
VA: Health and Benefits	☐	☐	☐

Sandboxx	☐	☐	☐
VET Tv	☐	☐	☐
Aim High Air Force	☐	☐	☐
Armed Forces Bank	☐	☐	☐
ASVAB Flashcards	☐	☐	☐
ASVAB Practice for Dummies	☐	☐	☐
ASVAB Practice Test 2024 Prep	☐	☐	☐
CBT-i Coach	☐	☐	☐
Defense & Military News	☐	☐	☐

End of Block: Perception of Data Sensitivity

Start of Block: Inferences and Attack Scenarios

Q22 Imagine that a malicious app collects the following data points/permissions from you and provides them to a threat actor:

Contact Info: Email Address, Name, Phone Number, Address

Identity Info: DOB, Age, Gender, Sexual Orientation

Military Info: Military Base/Installation, Branch

Financial Info: Financial Information

Device/App Info: Device Identifiers, App Activity

Sensor Permissions: Access Camera, Record Audio, Access Health and Fitness Data

Location Permissions: Access Location, Access Background Location

Communication Permissions: Make/Manage Phone Calls, Access Contacts, Access SMS/iMessages

These data points/permissions are derived from data flows we found in real apps marketing themselves toward military-affiliated personnel.

Q23 What pieces of additional information could a threat actor infer from those data points? List as many as possible. Where applicable, identify the data points used to derive each inference. For example, if a threat actor has address + location + financial information, they might be able to infer that the military member is living off base and has financial issues.

Q24 If the aforementioned data points/inferences were available to a threat actor, which of the following areas do you think could be impacted? Select all that apply.

- ☐ Personal Safety
- ☐ Security Clearance Eligibility
- ☐ Operational/Unit Security
- ☐ National Security
- ☐ None of the Above

Q25 For each area of impact you selected, describe attack or privacy-breach scenarios that could arise if a threat actor had the aforementioned data points/inferences.

	Scenarios
Personal Safety	
Security Clearance Eligibility	
Operational/Unit Security	
National Security	

End of Block: Inferences and Attack Scenarios

Start of Block: Mitigations

Q26 What privacy practices do you follow when using your device(s)? For each privacy practice, indicate whether you follow it on your personal device, government/military-issued device, or both.

Q27 Do you receive any security/privacy guidelines from the government/military on how to use your device(s)?

☐ Yes

☐ No

Q28 From whom do you receive these guidelines?

Q29 Briefly describe the guidelines you receive, including their names if applicable.

Q30 To what extent do you follow these guidelines?

- ☐ Entirely
- ☐ Mostly
- ☐ Somewhat
- ☐ A little
- ☐ Not at all
- ☐ Prefer not to say

Q31 In light of the aforementioned attack/privacy breach scenarios, to what extent do you think your privacy practices and the guidelines you receive reduce your privacy risks?

- ☐ Entirely
 - ☐ Mostly
 - ☐ Somewhat
 - ☐ A little
 - ☐ Not at all
-

Q32 In light of the aforementioned attack/privacy breach scenarios, to what extent do you think your privacy practices reduce your privacy risks?

- ☐ Entirely
 - ☐ Mostly
 - ☐ Somewhat
 - ☐ A little
 - ☐ Not at all
-

Q33 How effective would each of the following be in mitigating the aforementioned attack/privacy breach scenarios?

	Extremely effective	Very effective	Moderately effective	Slightly effective	Not effective at all
DoD-maintained "approved app list" for military-affiliated personnel to use on personal devices	☐	☐	☐	☐	☐
DoD extension of security/privacy standards for mobile apps marketed to military-affiliated personnel	☐	☐	☐	☐	☐
App stores setting and enforcing rules for mobile apps marketed to military-affiliated personnel	☐	☐	☐	☐	☐
U.S. federal law restricting data brokers from buying/selling data about military-affiliated personnel	☐	☐	☐	☐	☐
U.S. federal law requiring independent audits to verify privacy disclosures for mobile apps marketed to	☐	☐	☐	☐	☐

military-
affiliated
personnel

Stricter bans
on foreign
code in apps
marketed to
military-
affiliated
personnel

In-phone
warnings when
foreign or
unknown third-
party code is
present



Q34 How likely would you be to adopt or support the following mitigations?

	Extremely likely	Somewhat likely	Neither likely nor unlikely	Somewhat unlikely	Extremely unlikely
DoD-maintained “approved app list” for military- affiliated personnel to use on personal devices	☐	☐	☐	☐	☐
DoD extension of security/privacy standards for mobile apps marketed to military-affiliated personnel	☐	☐	☐	☐	☐
App stores setting and enforcing rules for mobile apps marketed to military-affiliated personnel	☐	☐	☐	☐	☐
U.S. federal law restricting data brokers from buying/selling data about military-affiliated personnel	☐	☐	☐	☐	☐
U.S. federal law requiring independent audits to verify privacy disclosures for mobile apps marketed to military-affiliated personnel	☐	☐	☐	☐	☐
Stricter bans on foreign/unknown SDKs	☐	☐	☐	☐	☐

In-phone
warnings when
foreign/unknown
SDKs are
present



End of Block: Mitigations

Start of Block: Demographics

Q35 Which of the following describes you? Select all that apply.

☐

U.S. Active Duty

☐

U.S. Reserve or National Guard Member

☐

U.S. Military Veteran

☐

Family Member of U.S. Active Duty / U.S. Reserve or National Guard Member /
U.S. Military Veteran (Please specify which and your relationship)

☐

U.S. DoD Civilian Employee

Q36 Which branch of the U.S. military are you most closely affiliated with?

- ☐ Air Force
 - ☐ Army
 - ☐ Coast Guard
 - ☐ Marine Corps
 - ☐ Navy
 - ☐ Space Force
 - ☐ Prefer not to say
-

Q37 What is your gender?

- ☐ Male
 - ☐ Female
 - ☐ Other (Please Specify) _____
 - ☐ Prefer not to say
-

Q38 What is your race/ethnicity?

- ☐ White
 - ☐ Black or African American
 - ☐ American Indian or Alaska Native
 - ☐ Asian
 - ☐ Native Hawaiian or Pacific Islander
 - ☐ Other (Please Specify) _____
 - ☐ Prefer not to say
-

Q39 What is your age range?

- ☐ 18-24
- ☐ 25-34
- ☐ 35-44
- ☐ 45-54
- ☐ 55-64
- ☐ 65+
- ☐ Prefer not to say

End of Block: Demographics
